$(10011.1011)_2 = (19.6875)_{10}$

16 ~~*~~ 2 1 0.5 + 0 ~~*~~ + 0.125 + 0.0625

$\boxed{19 \quad .6875}$

38.6875

19.6875

20.25

$(69.3125)_{10}$

$\Downarrow$

$(\quad\quad)_2$

1000101.0101 $\boxed{0.3125}$

0 ~~*~~ 0.25 ↑ 0.0625

1000101 . 0 1 0 0

~~*~~ 0.25 ~~*~~

1000101.1010

~~*~~ ~~*~~ ~~*~~ ~~*~~ ~~*~~ ~~*~~ ~~*~~ ~~*~~ 1

0.5 0 ~~*~~ +125 0 ~~*~~ 25

0.6125

$(0.625)_{10} = (0.101)_2$

$0.625 * 2 \rightarrow \boxed{1}.25 \rightarrow 1$

$0.25 * 2 \rightarrow \boxed{0}.5 \rightarrow 0$

$0.5 * 2 \rightarrow \boxed{1}.0 \rightarrow 1$

$\boxed{0}$

$(0.125)_{10} = (0.001)_2$

~~*~~ 0.5 ~~*~~ +125

$0.125 * 2 = 0.25 \rightarrow 0$

$0.25 * 2 = 0.5 \rightarrow 0$

$0.5 * 2 = 1.0 \rightarrow 1$

$\boxed{0}$

$(0.101)_2 = (0.625)_{10}$

$2^{-1}\ 2^{0}$ $2^{1}\ 2^{2}$

(0.101)

2^{-1} 2^{-2} 2^{-3}

0.5 0.25 0.125

0.5 + 0 + 0.125

$\boxed{0.625}$

$(0.0011)_2 = (\quad\quad)_{10}$

0.1875

0.0011

~~*~~ ~~*~~ ~~*~~ ~~*~~

2^{-1} 2^{-2} 2^{-3} 2^{-4}

0.125+0.0625

$\boxed{0.1875}$

Binary Numbering system

- → 0,1 , Base: 2
- → $(__)_2 = (?)_{10}$
- → $(__)_{10} = (?)_2$

- → $(0.__)_2 = (0.__)_{10}$ ✓
- → $(0.__)_{10} = (0.__)_2$ ✓
- → Addition $(__)_2 + (__)_2$
- → Subtraction $(__)_2 - (__)_2$
- → 1st & 2nd Complement

CamScanner

$(10011.1011)_2 = (19.6875)_{10}$

16 ~~8~~ ~~4~~ 2 1 0.5+0 ~~+0.25~~ +0.125 +0.0625

19 6875

38.6875

19.6875

20.25

(1 0 0 0 1 0 1)

~~128~~ 64 32 16 8 4 2 1

69
-64
5
-4
1
-1
0

$(69.3125)_{10}$

⇓

$(1000101.0101)_2$

0.3125 * 2 = 0.625 → 0

0.625 * 2 = 1.25 → 1

0.25 * 2 = 0.5 → 0

0.5 * 2 = 1.0 → 1

0

$(0.625)_{10} = (0.101)_2$

0.625 * 2 → 1.25 → 1

0.25 * 2 → 0.5 → 0

0.5 * 2 → 1.0 → 1

0

$(0.125)_{10} = (0.001)_2$

~~0.5~~ ~~0.25~~ ~~0.125~~

0.125 * 2 = 0.25 → 0

0.25 * 2 = 0.5 → 0

0.5 * 2 = 1.0 → 1

0

$(0.101)_2 = (0.625)_{10}$

2^{-1} 2^{-2} 2^{-3}

0.5 0.25 0.125

0.5 + 0 + 0.125

0.625

$(0.0011)_2 = (\quad)_{10}$

0.1875

0.0011

~~2^{-1}~~ ~~2^{-2}~~ 2^{-3} 2^{-4}

0.125+0.0625

0.1875

Binary Numbering system

- 0,1 , Base: 2
- $(__)_2 = (?)_{10}$
- $(__)_{10} = (?)_2$
- $(0.__)_2 = (0.__)_{10}$ ✓
- $(0.__)_{10} = (0.__)_2$ ✓
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- 1st & 2nd Complement

CamScanner

111

10101 → $(21)_{10}$

+ 1110 → $(14)_{10}$

100011 → $(35)_{10}$

32 ✗ ✗ ✗ 2 1

35

1 1 1 1 1

1101111 → $(55)_{10}$

+ 1111 → $(15)_{10}$

1000110 → 70

64 ✗ ✗ ✗ 4 2 ✗

1000110

100100000

101011

70

2	70	
2	35	0
2	17	1
2	8	1
2	4	0
2	2	0
2	1	0
	0	1

128 64 32 16 8 4 2 1

1 0 0 0 1 1 0

70
−64
6
−4
2
−2
0

1 1 1

10101
+ 1110
100011

28 + 5 = 33

13 + 5 = 18

0	0	1	1
0	1	0	1
0	1	1	10

$(1)_{10}$
$(1)_{10}$
$(2)_{10}$
$(10)_2$

Binary Numbering system

- 0,1 , Base: 2
- $(__)_2 = (?)_{10}$
- $(__)_{10} = (?)_2$
- $(0.__)_2 = (0.__)_{10}$ ✓
- $(0.__)_{10} = (0.__)_2$ ✓
- Addition $(__)_2 + (__)_2$ ✓
- Subtraction $(__)_2 - (__)_2$ ✓
- 1st & 2nd Complement

CamScanner

1000

0101

5

8 − 3 = 5

101000 + 11110 = 110110

54 − 14 = 40

101000

40

41 − 23 = 18

283

101001

10111

302 + 19 = 283

101001

10010 ✓

010110

10000

10100

22 − 9 = 13

18 − 7 = 11

0	1	1	10
0	0	1	1
0	1	0	1

Binary Numbering system

- 0,1 , Base: 2
- $(__)_2 = (?)_{10}$
- $(__)_{10} = (?)_2$
- $(0.__)_2 = (0.__)_{10}$ ✓
- $(0.__)_{10} = (0.__)_2$ ✓
- Addition $(__)_2 + (__)_2$ ✓
- Subtraction $(__)_2 - (__)_2$ ✓
- 1st & 2nd Complement ✓

CamScanner

$(10101)_2 =$ 1st ~~1~~010 2nd 1st 1010 + 1 = 1011

$(10010)_2 =$ 1st 1101 + 1 = 1110

Binary Numbering system

- → 0,1 , Base: 2
- → $(__)_2 = (?)_{10}$
- → $(__)_{10} = (?)_2$
- → $(0.__)_2 = (0.__)_{10}$
- → $(0.__)_{10} = (0.__)_2$
- → Addition $(__)_2 + (__)_2$
- → Subtraction $(__)_2 - (__$
- → 1st & 2nd Complement